Belgrade, Serbia

Nemanja Mikić

n0t_l3ss@outlook.com
linkedin.com/in/nrmikic
github.com/n0tl3ss

Education

- University of Belgrade, School of Electrical Engineering, Master's degree in Software Engineering (10/2017 - 09/2019)
- University of Belgrade, School of Electrical Engineering, Bachelor's degree in Software Engineering (10/2013 - 09/2017)

Employment

→ Oracle, Software Engineer (01/2022 - Present)

IC4 since 01/06/2024. Previously IC3 (01/01/2022 - 31/05/2024).

400+ merged pull requests across the Micronaut ecosystem since 2022.

Observability and framework integrations

- Implemented Micronaut OpenTelemetry support across HTTP and gRPC clients and servers, including span lifecycle handling, tracing annotations, SDK configuration, tests and documentation.
- Added JDBC/R2DBC tracing, Micrometer Observation and Prometheus PushGateway integrations, with configuration support, automated tests and documentation.

Cloud SDKs and service reliability

- Built AWS CloudWatch logging and OCI integrations for cloud logging, certificate management and reloading, OKE workload identity, and configurable signing-proxy requests.
- Fixed synchronous OCI retry backoff that could run on Netty event loops and stall server I/O. Added regression tests for responsive health checks, retry behavior and exception preservation.

Internal service frameworks and infrastructure

- Contributed to a Micronaut service framework for OCI services, adding raw metrics, dimensions and tags, HTTP/JVM and workflow-client instrumentation, metric-scope propagation and request-ID handling with automated tests.
- Extended Terraform reference environments with alarms, runtime deployment profiles, Redis networking, service-log forwarding and IAM policies.

Python interoperability and native developer tooling

- Improved GraalPy-based Python/Java interoperability in Micronaut, including coroutine/Publisher adaptation and automatic dataclass introspection with typed validation annotations.
- Extended Micronaut Python project generation with service discovery and Pyronaut tooling with async pytest, PyPI releases, multi-architecture containers and Windows PGO.

Selected published guides and sample applications

- Authored Micronaut Guides on Kubernetes (Oracle OKE, Amazon EKS, Google GKE) and metrics (AWS CloudWatch, Oracle Cloud). Co-authored OCI OpenTelemetry tracing.
- Added tested Python OAuth2/OIDC samples for Auth0, Keycloak and Amazon Cognito, plus CORS, token propagation, async HTTP clients and Oracle database integrations.

Delivery automation and integration testing

- Developed scheduled dependency-update and branch-synchronization tooling that creates review pull requests across Micronaut service and reference repositories. Implemented OCI resource-principal authentication for repository access.
- Implemented release pipelines with ticket checks, canary and template validation, environment tag promotion and rollback on failed validation. Added Maven/Gradle smoke-test jobs and shared dependency caches.

Multi-cloud application tooling

- Contributed to the GCN-to-GDK migration across BOMs, tooling, templates and CI. Expanded application/Kubernetes generation for AWS, OCI and GCP, with cloud configuration, metrics/tracing and generated tests.
- Implemented Oracle Functions generation and GitLab CI stages for JDK 17/21, GraalVM native builds, guide testing, dependency scanning and artifact publishing. Contributed recurring versioned releases.

→ Seven Bridges Genomics, Penetration Tester (05/2021 - 12/2021)

→ Seven Bridges Genomics, Software Engineer (12/2017 - 05/2021)

Core Team: developed platform infrastructure and backend services in Java, Python and Go.

- Built a secrets management service in Go with HashiCorp Vault, allowing users to store sensitive credentials and organization administrators to share secrets with authorized members.
- Implemented MFA recovery codes, email verification and time-based one-time passwords in a Python service. Improved OAuth2 single sign-on flows for custom applications using Java and Python.
- Helped split a large billing service into smaller Java services for customer charges and platform resource pricing. Developed an administrative billing API gateway and HTTP clients in Go.
- Implemented a Go service to monitor AWS EC2 test environments and shut them down after more than 90 minutes without user activity.
- Developed a Go and MongoDB messaging service for administrative announcements to anonymous users, registered users or individual recipients.
- Worked with Django, Jersey, Jackson, Guice, JDBI, Hibernate and go-chi across backend services and platform integrations.

→ Seven Bridges Genomics, Python Software Engineer Intern (03/2017 - 06/2017)

- Built a Python user management service for registration and authentication, and an API gateway for frontend clients using Tornado, SQLAlchemy and PostgreSQL. Worked with REST APIs, Git and a microservices architecture.

Technologies

Languages: Java, Python, Go. Frameworks and runtimes: Micronaut, Netty, GraalVM, GraalPy, Django, Tornado, Jersey.

Cloud and deployment: Oracle Cloud Infrastructure, AWS, Kubernetes, Docker, Terraform. Delivery: Git, Gradle, Maven, GitHub Actions, GitLab CI/CD.

Observability: OpenTelemetry, Micrometer, Prometheus, distributed tracing, metrics and logging.

Data and security: PostgreSQL, MongoDB, SQLAlchemy, Hibernate, JDBI, HashiCorp Vault, OAuth2, SSO and MFA.

Languages: Serbian, English